

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
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Help Button :	No
Show Reports :	No
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Corporate Finance

Section Id :	640653134555
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	23
Number of Questions to be attempted :	23
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653374469
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532040163 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : CORPORATE FINANCE

(COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536642686. ✓ Yes

6406536642687. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653374470

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406532040164 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A two-asset portfolio consists of 30% in Asset 1 ($r_1 = 0.10$, and $\sigma_1 = 0.2$) and 70% in Asset 2 ($r_2 = 0.25$, and $\sigma_2 = 0.4$). If the covariance between Asset 1 and Asset 2 is 0.1, then what is the expected return of the portfolio? (Note: r_i represents the expected rate of return and σ_i represents the standard deviation of the rate of return, respectively, for an asset i .)

Options :

6406536642688. ✗ 0.195

6406536642689. ✓ 0.205

6406536642690. ✗ 0.220

6406536642691. ✗ 0.235

Question Number : 3 Question Id : 6406532040165 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An equally weighted portfolio of 100 uncorrelated assets, each with a standard deviation of return $\sigma = 25$ %, has a portfolio standard deviation of

Options :

6406536642692. ✗ 250%

6406536642693. ✗ 25%

6406536642694. ✓ 2.5%

6406536642695. ✗ 0.25%

Question Number : 4 Question Id : 6406532040166 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

If the slope of the Capital Allocation Line (CAL) is 0.5 and the risk-free rate is 4%, what is the expected return of a portfolio on the CAL with a standard deviation of 12%?

Options :

6406536642696. ✖ 2%

6406536642697. ✖ 4%

6406536642698. ✖ 8%

6406536642699. ✔ 10%

Question Number : 5 Question Id : 6406532040167 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An individual with $u(c) = \ln(c)$ has a wealth of 1,000 units. There is a 10% chance of losing 500 due to an accident. What is the expected utility of this individual?

Options :

6406536642700. ✔ 6.84

6406536642701. ✖ 6.91

6406536642702. ✖ 6.21

6406536642703. ✖ 5.60

Question Number : 6 Question Id : 6406532040168 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A risk-neutral investor is offered a bet: 20% chance to win 500, 30% chance to win 200, and 50% chance to lose 100. What is the fair price of this bet?

Options :

6406536642704. ✖ 90

6406536642705. ✖ 100

6406536642706. ✔ 110

6406536642707. ✖ 120

Question Number : 7 Question Id : 6406532040169 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A risk-neutral lender requires an expected return of 10%. A borrower has a 20% probability of total default (0 recovery). What is the promised interest rate that the lender should charge?

Options :

6406536642708. ✖ 10%

6406536642709. ✖ 20%

6406536642710. ✖ 27.5%

6406536642711. ✔ 37.5%

Question Number : 8 Question Id : 6406532040170 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Asset A has $E(r) = 10\%$ and $\sigma = 15\%$. Asset B has $E(r) = 20\%$ and $\sigma = 25\%$. If the correlation between the assets $\rho_{AB} = 0$, what is the standard deviation of a portfolio with equal weights? (Note: r represents the rate of return and σ represents the standard deviation of the rate of return.)

Options :

6406536642712. ✔ 14.58%

6406536642713. ✖ 17.56%

6406536642714. ✖ 20.00%

6406536642715. ✖ 22.14%

Question Number : 9 Question Id : 6406532040171 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the Sharpe Ratio for a portfolio with an expected return of 15%, a risk-free rate of 3%, and a standard deviation of 20%?

Options :

6406536642716. ✖ 0.75

6406536642717. ✔ 0.60

6406536642718. ✖ 0.40

6406536642719. ✖ 0.20

Question Number : 10 Question Id : 6406532040172 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Given two assets with $\sigma_1 = 10\%$ and $\sigma_2 = 20\%$ and correlation $\rho = -1$. What is the weight of Asset 1 (w_1) that results in a zero-variance portfolio? (Note: σ_i represents the standard deviation of the rate of return for an Asset i)

Options :

6406536642720. ✖ 1.5

6406536642721. ✔ 0.67

6406536642722. ✖ 0.5

6406536642723. ✖ It is not possible to create a zero-variance portfolio.

Question Number : 11 Question Id : 6406532040173 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

The risk-free rate is 4%. The optimal risky portfolio has an expected rate of return $E(r_p) = 12\%$ and the standard deviation of the portfolio rate of return $\sigma_p = 15\%$. If an investor wants an expected return of 10%, what weight should she put in the risky portfolio?

Options :

6406536642724. ✖ 25%

6406536642725. ✖ 50%

6406536642726. ✔ 75%

6406536642727. ✖ 100%

Question Number : 12 Question Id : 6406532040174 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You short-sell 80 shares of a stock at Rs 60 per share. One year later, you buy them back at Rs 55 per share. Ignoring interest and transaction costs, what is your total profit?

Options :

6406536642728. ✖ Rs 480

6406536642729. ✔ Rs 400

6406536642730. ✖ Rs 320

6406536642731. ✖ You do not make any profit.

Question Number : 13 Question Id : 6406532040175 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A portfolio has a beta of 1.5. The risk-free rate is 5% and the market risk premium is 8%. According to CAPM, what is the expected return?

Options :

6406536642732. ✖ 5%

6406536642733. ✖ 10%

6406536642734. ✖ 12%

6406536642735. ✔ 17%

Question Number : 14 Question Id : 6406532040176 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

The risk-free rate is 4%. The optimal risky portfolio has an expected rate of return $E(r_p) = 12\%$ and the standard deviation of the portfolio rate of return $\sigma_p = 15\%$. If an investor invests 60% of his wealth in the optimal risky portfolio and the remaining wealth in a risk-free asset, what will be the standard deviation of the investor's portfolio?

Options :

- 6406536642736. ✖ 15%
- 6406536642737. ✖ 12%
- 6406536642738. ✔ 9%
- 6406536642739. ✖ 6%

Question Number : 15 Question Id : 6406532040177 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

There are two assets: Asset X ($r_x = 0.12$, $\sigma_x = 0.2$) and Asset Y ($r_y = 0.15$, $\sigma_y = 0.3$). The covariance between the two assets is $\sigma_{xy} = 0$. What is the Global Minimum Variance (GMV) weight for Asset X for the two-asset portfolio? (Note: r_i represents the expected rate of return and σ_i represents the standard deviation of the rate of return, respectively for an asset i .)

Options :

- 6406536642740. ✖ 30.77%
- 6406536642741. ✖ 40.00%
- 6406536642742. ✖ 60.00%
- 6406536642743. ✔ 69.23%

Question Number : 16 Question Id : 6406532040178 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

If an investor's risk aversion coefficient $A = 4$, the risk-free rate is 0.05, and the risky asset has an expected rate of return $E(r) = 0.13$ and the variance of rate of return $\sigma^2 = 0.04$. Assuming mean-variance preferences ($U = E(r) - \frac{1}{2}A\sigma^2$), what is the optimal weight (w^*) in the risky asset?

Options :

- 6406536642744. ✖ 25%
- 6406536642745. ✔ 50%
- 6406536642746. ✖ 75%
- 6406536642747. ✖ 100%

Question Number : 17 Question Id : 6406532040179 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A project pays 200 units with probability 0.25, 400 units with probability 0.40 and 600 units with probability 0.35, in the next year. If the expected annual discount rate is 10%, what is the project's present value? (Rounded off to nearest integer)

Options :

6406536642748. ✓ 382 units

6406536642749. ✗ 400 units

6406536642750. ✗ 420 units

6406536642751. ✗ 462 units

Question Number : 18 Question Id : 6406532040180 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An asset has a return of 20% in a "Boom" condition (probability 0.3), 8% in a "Normal" condition (probability 0.4) and 2% in a "Bust" condition (probability 0.3). What is the standard deviation of the asset return?

Options :

6406536642752. ✗ 9.80%

6406536642753. ✓ 7.12%

6406536642754. ✗ 5.74%

6406536642755. ✗ 3.21%

Question Number : 19 Question Id : 6406532040181 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

MNO Bank has total assets of Rs 500 billion and a leverage ratio of 5. What is the value of the total liabilities of MNO Bank?

Options :

6406536642756. ✗ Rs 100 billion

6406536642757. ✗ Rs 300 billion

6406536642758. ✓ Rs 400 billion

6406536642759. ✗ Rs 500 billion

Question Number : 20 Question Id : 6406532040182 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An investor with mean-variance preferences $U = \mu - \frac{1}{2}A\sigma^2$ is indifferent between Asset A ($\mu = 0.175, \sigma^2 = 0.02$) and Asset B ($\mu = 0.15, \sigma^2 = 0.01$). What is the investor's coefficient of

risk aversion (A)? (Note: μ represents the expected rate of return and σ^2 represents the variance of the rate of return.)

Options :

6406536642760. ✖ 12.5

6406536642761. ✖ 10

6406536642762. ✖ 7.5

6406536642763. ✔ 5

Question Number : 21 Question Id : 6406532040183 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A venture capitalist has a utility function of the form $U(x) = \sqrt{x}$. She invests in a new firm with a 50% chance of failing. If the new firm succeeds, she expects to earn 256 million units. In the case of failure, she only gets 16 million units back. The venture capitalist has an option to purchase an insurance policy that pays her 240 million units if her investment fails. If she pays a price p for this policy, she would be sure to get $(256 - p)$ million units regardless of success or failure. What would be the largest price (p) that the venture capitalist would be willing to pay for such an insurance?

Options :

6406536642764. ✖ 120 million units

6406536642765. ✔ 156 million units

6406536642766. ✖ 180 million units

6406536642767. ✖ 240 million units

Sub-Section Number :

3

Sub-Section Id :

640653374471

Question Shuffling Allowed :

No

Question Id : 6406532040184 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (22 to 23)

Question Label : Comprehension

A pharmaceutical company start a new project to discover a drug. The project costs 500 thousand units, which is financed with a 400 thousand-unit loan and the remaining amount in equity. There is a 70% chance of a successful discovery, and in this case, the project generates a payoff of 900 thousand units in the next year. In the event of failure (30% chance), only 100 thousand units are recovered. The appropriate cost of capital is 10% per year.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 22 Question Id : 6406532040185 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What would be the appropriate promised rate of return that the creditor of the loan would demand?

Options :

6406536642768. ✖ 28.33%

6406536642769. ✖ 35.27%

6406536642770. ✔ 46.43%

6406536642771. ✖ 58.57%

Question Number : 23 Question Id : 6406532040186 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What is the expected payoff for the equity owner in the next year?

Options :

6406536642772. ✖ 110.33 thousand

6406536642773. ✖ 147.67 thousand

6406536642774. ✖ 180.19 thousand

6406536642775. ✔ 220.00 thousand

Sub-Section Number :

4

Sub-Section Id :

640653374472

Question Shuffling Allowed :

No

Question Id : 6406532040187 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (24 to 26)

Question Label : Comprehension

Suppose there are two assets, Asset 1 and Asset 2, with the following characteristics:

- Expected Rate of Return: $r_1 = 0.15$, $r_2 = 0.20$
- Standard Deviation: $\sigma_1 = 0.20$, $\sigma_2 = 0.30$
- Correlation: $\rho_{12} = 0.3$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 24 Question Id : 6406532040188 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What would be the standard deviation of the rate of return of the equally weighted portfolio?

Options :

6406536642776. ✓ 0.2037

6406536642777. ✗ 0.2243

6406536642778. ✗ 0.2500

6406536642779. ✗ 0.2734

Question Number : 25 Question Id : 6406532040189 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Suppose you would like to construct a portfolio with an expected rate of return of 19%. What weights would you put on Asset 1 and Asset 2, respectively?

Options :

6406536642780. ✓ 20%, 80%

6406536642781. ✗ 40%, 60%

6406536642782. ✗ 60%, 40%

6406536642783. ✗ 80%, 20%

Question Number : 26 Question Id : 6406532040190 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

What are the weights of Asset 1 and Asset 2 in the minimum variance portfolio?

Options :

6406536642784. ✗ 83.2%, 16.8%

6406536642785. ✓ 76.6%, 23.4%

6406536642786. ✗ 62.4%, 37.6%

6406536642787. ✗ 48.3%, 51.7%